

Microphone & Audio CODEC Fundamentals

A complete, crystal-clear visual guide to acoustic sound capture, pre-amps, Delta-Sigma ADCs, digital filtering, and I2S forwarding using the ES8311 and ESP32-S3.

Target CODEC: **Everest Semi ES8311 (QFN-20)** Host Controller: **ESP32-S3 (I2S DMA + I2C)** Audio Quality: **24-Bit / 100 dB SNR**

1. Core Fundamentals: How Microphones Capture Sound

Sound is mechanical vibration travelling through air as pressure waves. A **microphone** is an acoustic-to-electrical transducer that converts microscopic sound pressure changes into electrical signals.

🔊 Analog Microphones (Electret ECM & Analog MEMS)

An analog microphone outputs a continuous AC voltage waveform. Because microphone diaphragms are microscopic, this raw output is tiny—typically between **1 mV and 10 mV RMS**. Without amplification, it is completely inaudible to a microcontroller.

Challenge: Weak analog signals easily pick up electrical noise from nearby WiFi antennas, power rails, and switching regulators.

⚡ Digital PDM Microphones (Digital MEMS)

A digital MEMS microphone integrates the sensor, pre-amp, and an internal 1-bit Delta-Sigma modulator onto a single silicon die. Instead of analog voltage, it outputs a high-frequency **Pulse Density Modulation (PDM)** digital bitstream.

Benefit: Highly immune to PCB trace noise and electromagnetic interference (EMI).

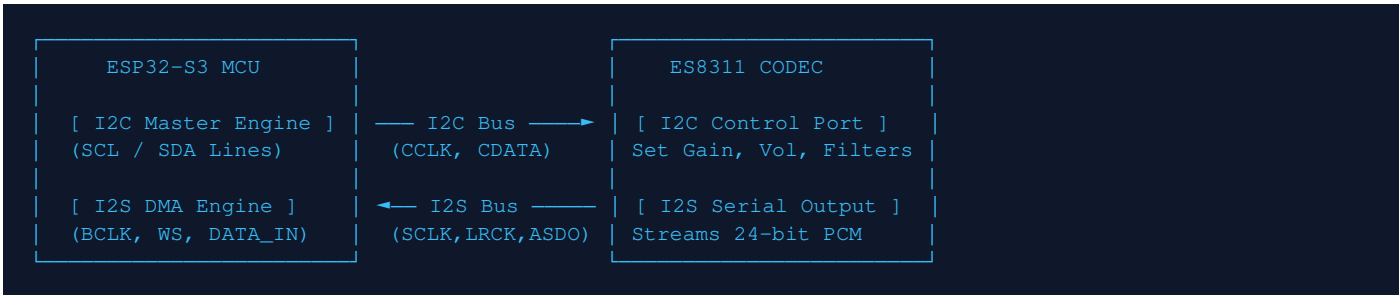
2. Why Do We Need the ES8311 Audio CODEC?

Connecting a microphone directly to general-purpose microcontroller ADC pins yields noisy, distorted audio. The **ES8311 Audio CODEC** solves all front-end challenges in hardware:

Feature	Microcontroller Built-In ADC (e.g. SAR)	ES8311 Dedicated Audio CODEC
Resolution	12-Bit (High quantization noise)	24-Bit Multi-Bit Delta-Sigma (Studio Grade)
Signal-to-Noise (SNR)	~55 dB to 65 dB (Audible hiss)	100 dB SNR (Crystal Clear)
PGA Pre-Amplifier	None (Requires external op-amps)	Integrated 0 dB to +30 dB PGA in 3 dB steps
DC Offset & Hum	Suffers from DC drift and supply ripple	Integrated 2-Stage Dynamic HPF
Dynamic Control	Must be coded in DSP software	Hardware Auto Level Control (ALC) + Noise Gate
CPU Overhead	High (Constant interrupt handling)	Zero (Streams via I2S DMA automatically)

3. Dual-Plane System Architecture: I2C Control vs I2S Data

The ESP32-S3 communicates with the ES8311 using two independent, dedicated bus interfaces:



- **Control Plane (I2C):** Low-speed configuration bus (100 kHz - 400 kHz). Used once during startup to set volume, pre-amp gain, filter modes, and clock dividers.
- **Data Plane (I2S):** High-speed audio streaming bus (2.8 MHz - 12.288 MHz). Streams audio data continuously into ESP32 DMA memory with zero CPU intervention.

4. The 6-Stage Audio Processing Pipeline

When sound enters the ES8311, it passes through 6 sequential hardware processing stages:



5. Pinout & Pin Role Directory (QFN-20 Package)

Pin #	Pin Name	Type	Hardware Description & Connection Guide
1	CCLK	Input	I2C Clock: Connect to ESP32 I2C SCL with a 3.3 kΩ pull-up resistor.
2	MCLK	Input	Master Clock: Sourced from ESP32 ($256 \times F_s = 12.288 \text{ MHz @ } 48 \text{ kHz}$).
3	PVDD	Power	I/O Pad Power Rail: Connect to 3.3V with 0.1 μF bypass cap.
4	DVDD	Power	Digital Core Power Rail: Connect to 1.8V or 3.3V with 0.1 μF cap.
5	DGND	Ground	Digital Ground: Tie to main system ground plane.
6	SCLK / DMIC_SCL	I/O	Dual Role: I2S Bit Clock input (slave mode) OR DMIC Clock output for PDM mic.
7	ASDOUT	Output	ADC Audio Out: Streams 24-bit serialized PCM data directly into ESP32 I2S In.
8	LRCK	I/O	Frame Clock / Word Select (WS): Left/Right channel toggle ($= F_s = 48 \text{ kHz}$).
9	DSDIN	Input	DAC Playback In: Tie to DGND in recording/mic-only projects.
10	AGND	Ground	Analog Ground: Quiet ground reference for analog input stages.
11	AVDD	Power	Analog Power Rail: 3.3V clean analog power supply with 1 μF 0.1 μF caps.
12, 13	OUTP, OUTN	Output	Speaker / Headphone: Leave unconnected (NC) in mic-only projects.
14	DACVREF	Analog Ref	DAC Voltage Ref: Connect 1 μF bypass capacitor to AGND.
15	ADCVREF	Analog Ref	ADC Voltage Ref: Connect 1 μF bypass capacitor to AGND. Crucial for SNR!
16	VMID	Analog Ref	Mid-Rail DC Bias (AVDD/2): Connect 1 μF cap to AGND. Do not float!
17	MIC1N	Input	Mic Negative Input: Inverting terminal of analog pre-amp (differential mode).
18	MIC1P / DMIC_SDA	Input	Dual Role: Mic Positive Input (analog mode) OR PDM Data In (digital mic mode).
19	CDATA	I/O	I2C Data Line: Connect to ESP32 I2C SDA with 3.3 kΩ pull-up resistor.
20	CE	Input	I2C Address Select: Tie to GND for address 0x18; tie to 3.3V for address 0x19.
21 (EP)	Exposed Pad	Ground	Thermal & Ground Pad: Solder bottom center pad directly to PCB Ground plane.

6. The 3 Hardware Wiring Schemes

Scheme 1: Differential Analog Mic (Best Noise Immunity)

MIC (+) —|—[1uF]—> Pin 18 (MIC1P) | MIC (-) —|—[1uF]—> Pin 17 (MIC1N) | GND —> AGND

* Register: Reg 0x14 = 0x16 (Differential Pair Selected, +18dB PGA Gain)

Scheme 2: Single-Ended Analog Mic (2-Wire ECM)

Signal —|—[1uF]—> Pin 18 (MIC1P) | AGND —|—[1uF]—> Pin 17 (MIC1N)

* Register: Reg 0x14 = 0x16, Reg 0x0E = 0x00

Scheme 3: Digital PDM MEMS Mic

CLK ← Pin 6 (DMIC_SCL) | DATA → Pin 18 (DMIC_SDA) | VDD ← 3.3V | GND → DGND

* Register: Reg 0x14[6] = 1 (DMIC_ON), Reg 0x0E = 0x60 (Power Down Analog Preamp)

7. 5-Step ESP32-S3 Software Bring-Up Recipe

To start recording 24-bit audio at 48 kHz, send this exact sequence over I2C during system boot:

```
// 1. Soft Reset & Start Reference Voltage Pre-charge
i2c_write(0xFA, 0x01); // Soft reset all registers to default
delay_ms(10);
i2c_write(0x0D, 0x01); // Start VMID fast pre-charge
delay_ms(20); // Wait for VMID bypass capacitor to stabilize
i2c_write(0x0D, 0x02); // Switch VMID to normal low-noise operating state

// 2. Configure System Clocks (Slave Mode, MCLK from ESP32)
i2c_write(0x00, 0x80); // Enable State Machine in slave mode
i2c_write(0x01, 0x3F); // Turn ON MCLK, BCLK, and ADC digital/analog clocks
i2c_write(0x03, 0x10); // Single speed mode (8-48 kHz), OSR = 64x

// 3. Configure Serial Data Port Output (I2S Format, 24-bit depth)
i2c_write(0x0A, 0x00); // I2S Standard format, 24-bit word length, unmuted

// 4. Set Analog Mic Routing, PGA Gain (+18dB) & High-Pass Filter
i2c_write(0x14, 0x16); // Differential Mic mode, PGA Gain = +18 dB
i2c_write(0x1C, 0x60); // Enable Dynamic High-Pass Filter (removes DC offset)
i2c_write(0x17, 0xBF); // ADC Digital Volume = 0 dB (Unity Gain)

// 5. Power Up Analog Front-End & Start Streaming
i2c_write(0x0E, 0x00); // Power up Analog PGA and Delta-Sigma Modulator
i2c_write(0x44, 0x00); // Duplicate mono mic onto both Left & Right I2S slots
```

8. Troubleshooting & Cheat Sheet

Symptom	Root Cause	Hardware / Software Solution
Dead Silence / All Zeroes	Wrong I2C Address or missing MCLK	Verify Pin 20 (CE) is GND (0x18) or 3.3V (0x19). Ensure ESP32 outputs MCLK clock (256 × Fs).
Audio Very Faint / Quiet	PGA Gain too low	Increase PGA Gain in Register 0x14 [3:0] to 0x8 (+24 dB) or 0xA (+30 dB).
Loud WiFi / 50Hz Buzz	Missing VMID cap or single-ended pickup	Ensure 1 μF capacitor is present on Pin 16 (VMID). Use differential wiring for microphone.
Distorted / Clipped Audio	Input signal overdriving PGA	Lower PGA Gain in Reg 0x14 to +6 dB, or enable Auto Level Control (ALC) in Reg 0x18 (0x80).